

Ethernet & Bluetooth



School of Computer Science Engineering and Technology
Bennett University
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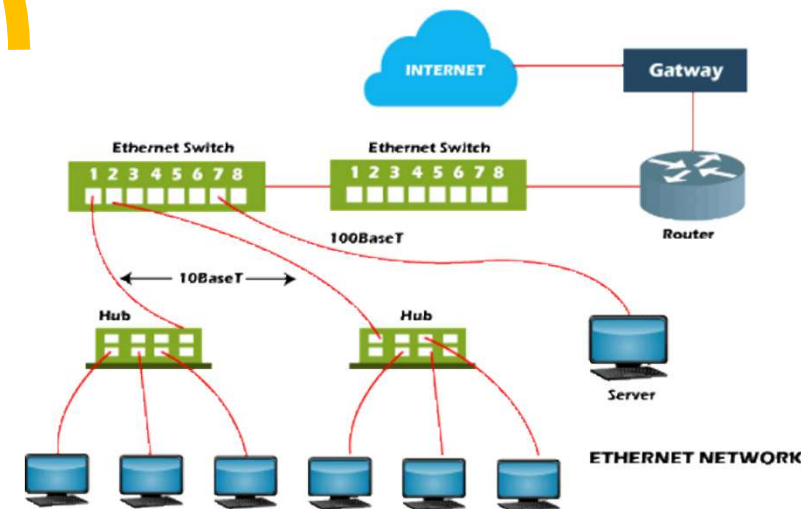
Outline

Ethernet

Bluetooth

Ethernet : Definition

- Ethernet is a type of communication technology that is created at **Xerox PARC in 1973 by Robert Metcalfe**.
- Ethernet is used to connect several computers to make a LAN.
- It is a widely used LAN protocol that connects computers within the LAN and WAN.
- A simple user interface to connect various devices easily.
- Devices can be connected using Ethernet port.
- The wireless networks replaced Ethernet in many areas; however, Ethernet is still more common for wired networking.
- Wired connections are **more secure and are less prone to interference**.

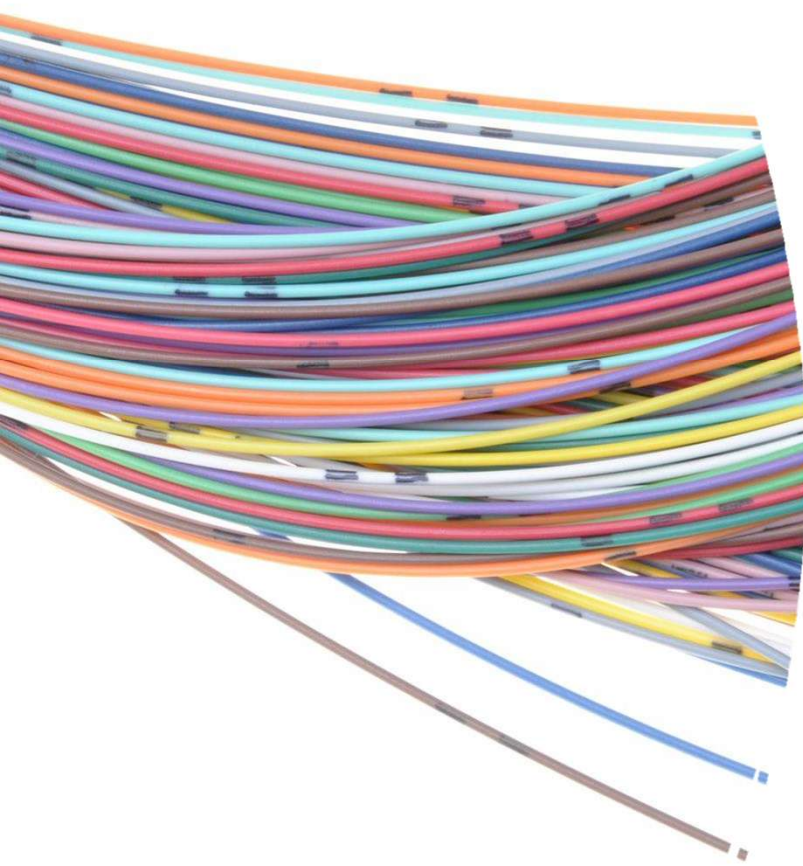


Ethernet : Features

- It is used by specific organizations for local networks, organizations such as school campuses and hospitals, company offices, etc. for its high speed, security, and reliability.
- Ethernet is of **low price**.
- Ethernet delivers higher levels of performance with maintaining **backward compatibility**.

Ethernet : Types

- **Fast Ethernet:** Supported by a twisted pair or CAT5 cable, can transfer or receive data at around 100 Mbps. The 100BASE-TX, 100BASE-FX, and 100BASE-T4 are the three categories of Fast Ethernet.
- **Gigabit Ethernet:** It uses fiber optic cable and twisted pair cable to create communication and it can transfer data at a rate of 1000 Mbps or 1Gbps.
- **10-Gigabit Ethernet:** The network can transmit data at a rate of 10 Gigabit/second, uses CAT6a or CAT7 twisted-pair cables and fiber optic cables as well. This network can be expended up to nearly 10,000 meters with the help of using a fiber optic cable.
- **Switch Ethernet:**
 - The network involves adding switches or hubs, which helps to improve network throughput as each workstation can have its own dedicated 10 Mbps connection instead of sharing the medium.
 - Instead of using a crossover cable, a regular network cable is used when a switch is used in a network.
 - The main task of the switch in a network is to transfer the data from one device to another device in the same network without affecting the other devices.
 - It supports 1000Mbps to 10 Gbps for latest Ethernet and 10Mbps to 100Mbps for fast Ethernet. This type of Ethernet makes use of star topology.



Ethernet : Types II

Various Types of Ethernet Cables

- 10Base2: This is a thin twisted pair coaxial cable.
 - 10Base5: This is thick twisted pair coaxial cables.
 - 10Base T: This is a twisted pair cable which offers a speed of around 10 Mbps.
 - 100BaseTX: This is a twisted pair cable and offer a speed of 100 Mbps.
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- 100Base FX: Fiber optic protocol which offers a speed of 100 Mbps.
 - 1000Base SX: Fiber optic protocol which utilizes a wavelength of 850nm for multimode networks.
 - 1000Base LX: Fiber optic protocol which utilizes a wavelength of 1310 nm, for multimode networks and up to 1550nm for singlemode networks.

Ethernet : Advantage & Disadvantage

Advantage



It is relatively inexpensive.



It provides high security for data as it uses firewalls in terms of data security.



It can transmit data at a speed of 1-100Gbps.



The quality of the data transfer does maintain.



Administration and maintenance are easier.

Disadvantage

- It needs deterministic service; therefore, it is not considered the best for real-time applications.
- Restricts in terms of distances.
- If you create a wired ethernet network that needs cables, hubs, switches, routers, they increase the cost of installation.
- Data needs quick transfer in an interactive application.
- Acknowledge is not sent by receiver after accepting a packet.
- It is not easy to determine which node or cable is causing the problem.

Bluetooth

- It is a cable replacement technology
- When two Bluetooth devices come within **50m range** of each other, they establish connection together
- Uses **radio waves** for communication
- Ability to **communicate with many different devices** at once without interference
- Open standard for short-range transmission of digital voice and data that supports point-to-point and multipoint applications
- Operates
 - in the unlicensed ISM band at 2.4 GHz which is available globally, along slight variation of location and width of band apply
 - Frequency Hopping scheme (1600 hops/sec)
 - 1 Mb/s symbol rate
 - Range 10+ meters

Bluetooth (Cont...)

- Bluetooth supports
 - Synchronous & asynchronous data channels.
 - Three simultaneous synchronous voice channels, or
 - One channel, with asynchronous data and synchronous voice
 - Each voice channel supports 64 kb/s in each direction.
 - The channel can support maximal 723.2 kb/s asymmetric (and still up to 57.6 kb/s in the return direction), or 433.9 kb/s symmetric.
- Bluetooth provides
 - point-to-point connection (only two Bluetooth units involved), or
 - point-to-multipoint connection.
- Support both packet-switching and circuit-switching

Bluetooth (Cont...)

- Bluetooth Special Interest Group (SIG) founded in Spring 1998 by Ericsson, Intel, IBM, Nokia, Toshiba;
- Now more than 2000 organizations joint the SIG
- Key features:
 - Robustness
 - low complexity
 - low power consumption, which enables maximum performance longevity for battery powered devices
 - low cost
- Limitations:
 - Security
 - Interference with wireless LANs

Application: Synchronization

- Data Access Points
- Synchronization
- Headset
- Conference Table
- Cordless Computer
- Business Card Exchange
- Instant Postcard
- Computer Speakerphone

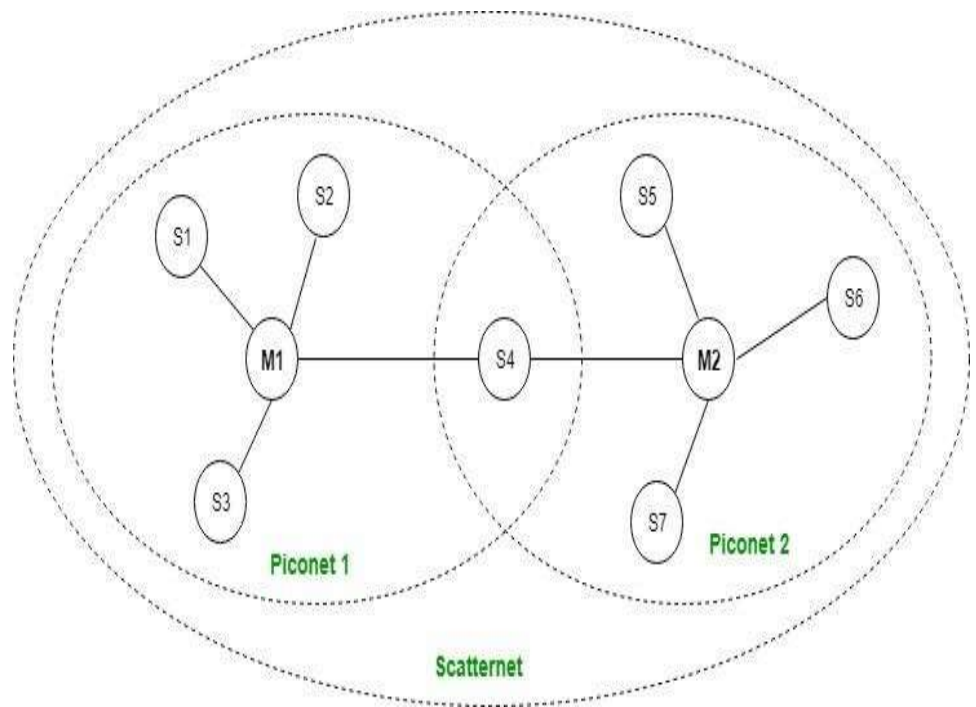


Bluetooth Topology

- Bluetooth-enabled devices can automatically locate each other
- Topology is established on a temporary and random basis
- Up to eight Bluetooth devices may be networked together in a **master-slave relationship to form a Piconet**
- Piconet is an ad-hoc computer network of devices using Bluetooth technology protocols to allow **one *master*** device to interconnect with up to **seven active *slave* devices**
- **Adhoc network:**
 - is a network connection method which is most often associated with wireless devices.
 - The connection is established for the duration of one session and requires **no base station**.
 - Instead, devices discover others within range to form a network for those computers.
 - Devices may search for target nodes that are out of range by flooding the network with broadcasts that are forwarded by each node.
 - Connections are possible over multiple nodes (multihop ad hoc network).
 - Routing protocols then provide stable connections even if nodes are moving around

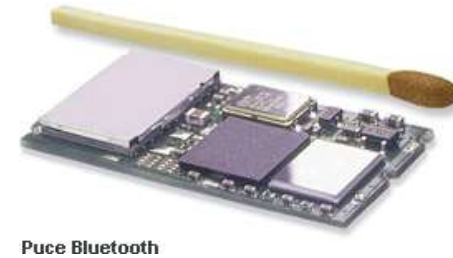
Bluetooth Topology (Cont ...)

- One is master, which controls and setup the network
- All devices operate on the same channel and follow the same frequency hopping sequence
- Two or more piconet interconnected to form a **scatter net**
- Only one master for each piconet
- A device can't be masters for two piconets
- The slave of one piconet can be the master of another piconet

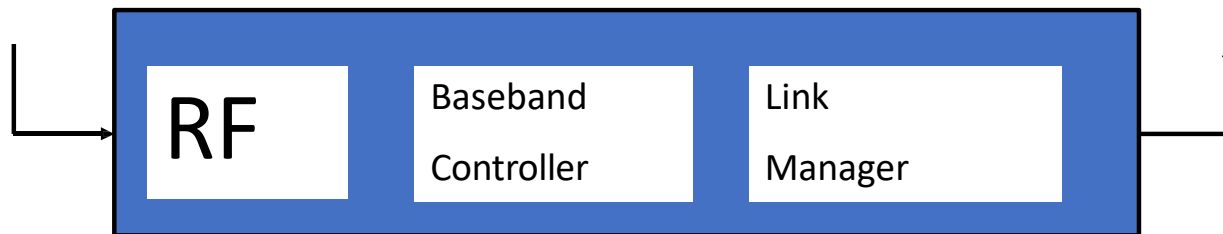


Bluetooth Working

- Bluetooth is a standard for tiny, radio frequency chips that can be plugged into your devices
- These chips were designed to take all of the information that your wires normally send, and transmit it at a special frequency to something called a receiver Bluetooth chip
- The information is then transmitted to your device



Bluetooth Chip



IEEE Standards

- Institute of Electrical and Electronics Engineers (IEEE) is an organization composed of engineers, scientists, and students which develops standards for the computer and electronics industry.
- Various standards are-
 - IEEE 802 LAN/MAN Overview and architecture
 - IEEE 802.1 LAN/MAN Bridging and management
 - IEEE 802.1x Port-based network access control
 - IEEE 802.2 Logical Link Control (LLC)
 - IEEE 802.3 Ethernet
 - IEEE 802.3ae 10 Gigabit Ethernet
 - IEEE 802.8 Fiber Optic
 - IEEE 802.11 Wi-Fi
 - IEEE 802.15 Wireless Personal Area Network (WPAN) e.g., bluetooth

References

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